

9.3 Direct Preference Optimization

Same pairs. Implicit $r = \beta \log(\pi/\pi_{\text{ref}})$. $Z(x)$ cancels. Offline.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Direct Preference Optimization (DPO) (Rafailov et al., 2023) fits a policy to chosen/rejected pairs **without** training an RM and **without** PPO. The reward is written in closed form from the policy and a frozen reference, then the Bradley–Terry model becomes a classification loss on logits.

First time this method appears. DPO fits a policy to the same preference pairs **without** an RM loop and **without** PPO.

What. Under Bradley–Terry, the optimal reward is $r = \beta \log(\pi / \pi_{\text{ref}}) + c(x)$. Plug into the logistic loss; r disappears. Train π_θ offline. **Why.** RLHF’s PPO stage is heavy. DPO is a classification loss on logits of y_w and y_l . **Architecture.** Policy + frozen reference. No critic, no sampling-in-the-loop required (you need logprobs of both completions). **How.** Lab 9: toy scalar logits. Start: $\pi = \pi_{\text{ref}} \Rightarrow \text{loss } \log 2$.

Formula. $\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right)$. **Tradeoffs.** + Offline, no RM. – Still needs pairs and a reference; can overfit; not magic if the pairs are noisy.

1. Skip the reward model

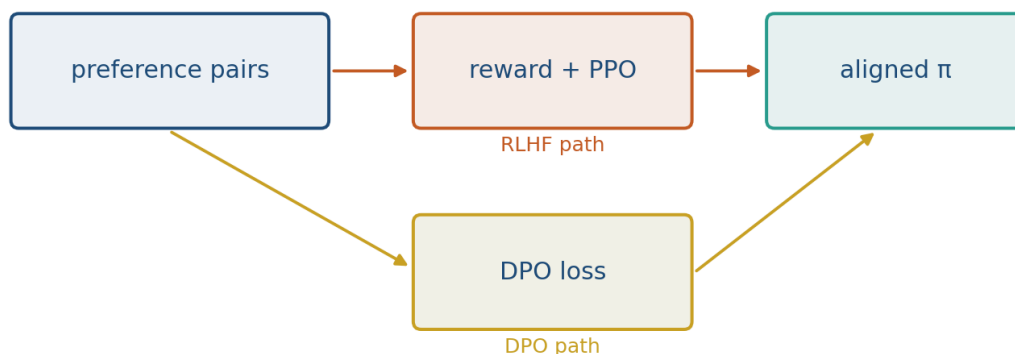
RLHF solves $\max_\pi \mathbb{E}[r] - \beta \text{KL}(\pi \| \pi_{\text{ref}})$. Under Bradley–Terry, the optimal r can be expressed as

$$r(x, y) = \beta \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} + \text{const}(x).$$

Plug that into the RM logistic loss and the r parameters disappear. You train π_θ directly.

Stanford CS224N 2025 L10 writes the same derived reward and stresses that the **partition** $Z(x)$ cancels because Bradley–Terry only sees a difference. KTO and IPO sit in the same family; this hour is DPO.

DPO fits the same pairwise data without an explicit reward model



2. The loss

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right).$$

In a batch you need log-probabilities of **both** completions under π_{θ} and under π_{ref} . The reference stays frozen (usually SFT). β is the same kind of leash as in RLHF: it scales how hard you push away from π_{ref} .

Lab 9 will compute this on toy scalar logits, not on an LLM.

If $\pi_{\theta} = \pi_{\text{ref}}$, both log-ratios are 0, the argument of σ is 0, and $\sigma(0) = 1 / 2$. Then $\mathcal{L}_{\text{DPO}} = \log 2 \approx 0.693$. That is the **start** of training, not a bug.

Write the argument of σ as $\beta (\Delta \log \pi_{\theta} - \Delta \log \pi_{\text{ref}})$ if it helps: you want π_{θ} to put relatively more mass on y_w than the reference did, compared with y_l . Absolute log-prob can fall on both sequences; **relative** log-prob is what DPO scores.

3. When DPO is the right picture

You have pairs, a GPU, and no appetite for a PPO stack. Offline: the pair set is fixed, so you do not explore new y unless you collect more. If the SFT model never proposed good y_w , DPO cannot invent them from nothing. Implicit reward can still overfit the pair set.

Report β , the reference checkpoint, and a held-out preference accuracy (how often π_{θ} assigns higher log-prob to y_w than to y_l). That number is not “the model is aligned”; it is “the model copied this rater.”

4. Teaching this note

About **35–40 minutes** at the board: write implicit r , plug into σ , compute $\sigma(0) = 1 / 2$. Play the Deep Dive **preference / RLHF** segment **2:48:26–3:09:39**. Lab 9 section 2 is the toy DPO scalar.

5. Worked example

Let $\beta = 1$. Suppose at init $\pi_{\theta} = \pi_{\text{ref}}$. Then

$$\log \frac{\pi_{\theta}(y_w)}{\pi_{\text{ref}}(y_w)} - \log \frac{\pi_{\theta}(y_l)}{\pi_{\text{ref}}(y_l)} = 0,$$

so $\mathcal{L} = -\log \sigma(0) = -\log(1 / 2) = \log 2$.

After a few steps, toy log-probs (sums over tokens): $\log \pi_{\theta}(y_w) = -2.0$, $\log \pi_{\text{ref}}(y_w) = -2.5$, $\log \pi_{\theta}(y_l) = -3.0$, $\log \pi_{\text{ref}}(y_l) = -2.0$. Log-ratio winner: $-2.0 - (-2.5) = 0.5$. Log-ratio loser: $-3.0 - (-2.0) = -1.0$. Difference $0.5 - (-1.0) = 1.5$. $\sigma(1.5) \approx 0.818$, loss $-\log 0.818 \approx 0.201$.

The frozen reference is still there: without it, π_{θ} can raise both log-probs or collapse. PPO can **sample new** y that were not in the pair file; offline DPO cannot.

 Init loss $\propto \log \frac{\pi_{\theta}(y_2)}{\pi_{\theta}(y_1)}$

log2, then a 1.5 logit gap" src="graphics/9.3-dpo/dpo-init.png" />

6. Where students get stuck

- Forgetting $\sigma(0) = 1/2$ when $\pi_{\theta} = \pi_{\text{ref}}$ at step 0.
 - Dropping π_{ref} because “there is no RM.” The reference **is** the leash.
 - Expecting DPO to invent replies the SFT model never wrote.
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7. Video

[Karpathy — Deep Dive into LLMs like ChatGPT](#). Play the **preference** segment **2:48:26–3:09:39** (RLHF). Pause on comparison data: DPO uses those same pairs, without the PPO loop. The closed-form loss is the board.

8. Practice

1. If $\pi_{\theta} = \pi_{\text{ref}}$, what is $\mathcal{L}_{\text{DPO}}$? (Compute $\sigma(0)$.)
2. Why do you still keep a frozen reference if there is no RM?
3. Name one thing PPO can do that offline DPO cannot, given a fixed pair file.
4. $\beta = 2$, winner log-ratio 0.4, loser log-ratio 0.1. What is the argument of σ , and is the loss smaller or larger than log2?
5. Both completions have log-prob -10 under π_{θ} and under π_{ref} . What is the DPO loss, and what did the policy fail to learn yet?